

Laboratory of Microeconomics I

Fundamentals of Real Analysis

Centro de Investigación y Docencia Económicas

Maestría en Economía

Arturo López

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Introduction

This document serves as a mathematical foundation review for Microeconomics I. It's designed to cover the essential mathematical tools we will need at the beginning of the semester. While it aims to be broad and comprehensive, it is by no means exhaustive. Most of the material presented here will be studied in greater depth during your Mathematics course.

The material draws heavily from [Lebl \(2023\)](#), [Dam \(2023\)](#), [Ok \(2007\)](#), [Kumaresan \(2005\)](#) and [De la Fuente \(2000\)](#).

For brevity, we will omit basic notions of set theory and functions. For a concrete review, see [De la Fuente \(2000\)](#), sections 1.1 and 1.4).

1 Binary Relations

Consider a set X . That is, a collection of objects which we call elements of X . The set X is just a collection of objects, with no additional structure whatsoever. We want to endow X with some structure. We will begin with the concept of a *binary relation*.

1.1 Definition and Examples

Definition 1.1 (Binary Relation). Let X and Y be two sets. A subset R of the Cartesian product $X \times Y$ is called a *binary relation* from X to Y . If $(x, y) \in R$, we write xRy .

Intuitively, the concept of a *binary relation* provides a way to formalize the idea that two objects, typically two elements of the same set, stand in a certain relationship to each other.

Example 1.1. Let $X = \{1, 2, 4\}$ and $Y = \{2, 3\}$. Then $R = \{(1, 2), (1, 3), (2, 3)\}$ defines a binary relation from X to Y that we might interpret as "<" (less than). Note that $(2, 4)$ is not a binary relation from X to Y (why?).

The following types of binary relations are of significant importance for microeconomic theory.

1.1.1 Preorderings

A class of binary relations of particular interest allows us to formalize the idea that some elements of a set "dominate" others in a specific sense. Such relations are called order relations, and a set endowed with an order relation is called an ordered set.

Definition 1.2 (Preorder). A binary relation $\succsim$ on a non-empty set X is called a *preorder* if it satisfies the following properties:

- (i) **Reflexivity:** $x \succsim x$ for all $x \in X$.
- (ii) **Transitivity:** If $x \succsim y$ and $y \succsim z$, then $x \succsim z$ for all $x, y, z \in X$.

The pair $(X, \succsim)$ is called a *preordered set*.

Definition 1.3 (Total Preorder). A preorder $\succsim$ on a non-empty set X is called a *total preorder* (or *complete preorder*) if it additionally satisfies:

- (iii) **Completeness:** For all $x, y \in X$, either $x \succsim y$ or $y \succsim x$ (or both).

The pair $(X, \succsim)$ is called a *totally preordered set*.

Intuitively, we interpret the relation $x \succsim y$ as " x dominates y " in some relevant sense (e.g., " x is at least as large as y ", " x is at least as preferred as y ", and so on).

Definition 1.4 (Equivalence Relation). A binary relation $\sim$ on a non-empty set X is called an *equivalence relation* if it satisfies the following properties:

- (i) **Reflexivity:** $x \sim x$ for all $x \in X$.
- (ii) **Symmetry:** If $x \sim y$, then $y \sim x$ for all $x, y \in X$.
- (iii) **Transitivity:** If $x \sim y$ and $y \sim z$, then $x \sim z$ for all $x, y, z \in X$.

For any $x \in X$, the *equivalence class* of x is defined as the set

$$[x]_{\sim} := \{y \in X \mid y \sim x\}.$$

An equivalence relation formalizes the notion of identifying two distinct objects when they share a particular property of interest. Naturally, such identification should satisfy certain consistency requirements. Intuitively, any object x must be considered identical to itself. If x is identified with y , then y must be identified with x . Finally, if x and y are regarded as identical, and y and z are also regarded as identical, then x should be considered identical to z . The properties of *reflexivity*, *symmetry*, and *transitivity* formalize these intuitive notions, respectively. A binary relation satisfying all these three properties is called an **equivalence relation**.

Example 1.2 (Consumer Preferences). Let $x \in X \subseteq \mathbb{R}_+^n$ represent the vector of quantities of n goods consumed by an agent, where $X \subseteq \mathbb{R}_+^n$ is called the **consumption set**. Define a **preference relation** $\succsim$ on X as follows: we read $x \succsim y$ as " x is at least as good as y ". From the preference relation $\succsim$, we can derive two other important relations on X :

- (a) The **strict preference relation**, $\succ$, defined by

$$x \succ y \iff x \succsim y \text{ and } y \not\succsim x$$

and read as " x is (strictly) preferred to y ".

- (b) The **indifference relation**, $\sim$, defined by

$$x \sim y \iff x \succsim y \text{ and } y \succsim x$$

and read as " x is indifferent to y ".

The relation $\succsim$ is called *rational*^a if it satisfies:

- (i) **Completeness:** For all $x, y \in X$, either $x \succsim y$ or $y \succsim x$ or both.
- (ii) **Transitivity:** For all $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

^aNote that a rational relation is a complete preorder.

The following result characterizes the properties of the derived relations when preferences are rational.

Lemma 1.1. *If $\succsim$ is rational, then the indifference relation $\sim$ is an equivalence relation, but the strict preference relation $\succ$ is not.*

Exercise 1.1. Prove **Lemma 1.1**.

2 Metric Spaces

In this section, we endow a set X with a different kind of structure that allows us to formalize the notion of the "distance" between two elements of X .

2.1 Definition and Examples

A *metric space* is a set equipped with a function that quantifies the distance between any two of its elements. This distance function (or *metric*) must satisfy certain natural and intuitive conditions that capture our geometric understanding of a distance between two points.

Definition 2.1 (Metric). Let X be a nonempty set. A function $d : X \times X \rightarrow \mathbb{R}$ is said to be a *metric* or a *distance function* on X if it satisfies the following properties:

- (i) **Non-negativity:** $d(x, y) \geq 0$ for all $x, y \in X$ and,
- (ii) **Identity of Indiscernibles:** $d(x, y) = 0$ iff $x = y$.
- (iii) **Symmetry:** $d(x, y) = d(y, x)$ for all $x, y \in X$.
- (iv) **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$.

The pair (X, d) is then called a *metric space*.

Example 2.1 (Standard Metric on $\mathbb{R}$). The most known example is the set $\mathbb{R}$ of real numbers endowed with the metric $d(x, y) := |x - y|$. This metric is induced by the absolute value function (which is the Euclidean norm in $\mathbb{R}$) and corresponds to our usual notion of distance on the real line.

Example 2.2 (Discrete Metric). Let X be a nonempty set. For any $x, y \in X$, define the *discrete metric* $d : X \times X \rightarrow \mathbb{R}$ by:

$$d(x, y) := \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases} \quad (2.1)$$

The pair (X, d) is a metric space.

The discrete metric turns any set into a metric space where distinct points are always at a unit distance from each other.

Example 2.3. Consider the set $\mathcal{C}([a, b])$ of all continuous functions $f : [a, b] \rightarrow \mathbb{R}$. For all $1 \leq p < \infty$, define the function $d_p : \mathcal{C}([a, b]) \times \mathcal{C}([a, b]) \rightarrow \mathbb{R}$ by:

$$d_p(f, g) := \left[\int_a^b |f(x) - g(x)|^p dx \right]^{1/p}, \quad \forall f, g \in \mathcal{C}([a, b]) \quad (2.2)$$

The pair $(\mathcal{C}([a, b]), d_p)$ is a metric space.

Moreover, define the function $d_\infty : \mathcal{C}([a, b]) \times \mathcal{C}([a, b]) \rightarrow \mathbb{R}$ by:

$$d_\infty(f, g) := \sup_{x \in [a, b]} |f(x) - g(x)|, \quad \forall f, g \in \mathcal{C}([a, b]) \quad (2.3)$$

The pair $(\mathcal{C}([a, b]), d_\infty)$ is a metric space.

Example 2.4. Let (X, d_X) and (Y, d_Y) be metric spaces. Then, for all $1 \leq p < \infty$, define the function $d : X \times Y \rightarrow \mathbb{R}$ by:

$$d((x_1, y_1), (x_2, y_2)) := [d_X(x_1, x_2)^p + d_Y(y_1, y_2)^p]^{1/p} \quad \forall (x_1, y_1), (x_2, y_2) \in X \times Y \quad (2.4)$$

The pair $(X \times Y, d)$ is a metric space.

We now introduce the *standard* or *Euclidean metric* on $\mathbb{R}^n$, which is the metric induced by the Euclidean norm on $\mathbb{R}^n$.

Recall: Every inner product $\langle \cdot, \cdot \rangle$ induces a norm via $\|x\| = \sqrt{\langle x, x \rangle}$. Moreover, every norm on a vector space induces a metric via $d(x, y) = \|x - y\|$. Therefore, from any inner product space $(X, \langle \cdot, \cdot \rangle)$, we can induce a normed vector space $(X, \|\cdot\|)$ and then induce a metric space (X, d) .

Example 2.5 (Euclidean Metric on $\mathbb{R}^n$). Consider $\mathbb{R}^n$ equipped with the standard inner product (dot product):

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

for $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbb{R}^n$.

This inner product induces the *Euclidean norm*:

$$\|x\|_2 = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^n x_i^2}$$

The pair $(\mathbb{R}^n, \|\cdot\|)$ is called the *n-dimensional Euclidean space*.

The Euclidean norm, in turn, induces the *Euclidean metric* on $\mathbb{R}^n$ defined by:

$$d_2(x, y) = \|x - y\|_2 = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

It is straightforward to verify that d_2 satisfies all the metric axioms, making $(\mathbb{R}^n, d_2)$ a metric space (for the triangle inequality, you may use the Minkowski's Inequality for ℓ^p norms).

Example 2.6 (p -Metrics on $\mathbb{R}^n$). Now that we understand how norms induce metrics, we can introduce a family of important metrics on $\mathbb{R}^n$. For each $1 \leq p \leq \infty$, $(\mathbb{R}^n, d_p)$ is a metric space, where $d_p : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}_+$ is defined by:

$$d_p(x, y) := \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}} \quad \text{for } 1 \leq p < \infty$$

and

$$d_\infty(x, y) := \max\{|x_i - y_i| : i = 1, \dots, n\} \quad \text{for } p = \infty$$

These metrics are induced by the corresponding ℓ^p norms $\|x\|_p := (\sum_{i=1}^n |x_i|^p)^{\frac{1}{p}}$ via $d_p(x, y) = \|x - y\|_p$.

Exercise 2.1. Let $X = \mathbb{R}$ and define $d : X \times X \rightarrow [0, \infty)$ by:

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ |x| + |y| & \text{if } x \neq y \end{cases}$$

Prove that d is a metric on $\mathbb{R}$.

(*Hint:* Verify the properties of a metric (**Definition 2.1**). For the triangle inequality, remember the triangle inequality of the euclidean norm: $|x + y| \leq |x| + |y|$.)

Exercise 2.2. Let (X, d_1) and (Y, d_2) be metric spaces and the function $d_3 : (X \times Y) \times (X \times Y) \rightarrow \mathbb{R}$ defined as follows:

$$d_3((x_1, y_1), (x_2, y_2)) = d_1(x_1, x_2) + d_2(y_1, y_2), \quad \forall (x_1, y_1), (x_2, y_2) \in X \times Y$$

Show that $(X \times Y, d_3)$ is a metric space.

Exercise 2.3. Let $X = \mathbb{R}$ and define $d : X \times X \rightarrow [0, \infty)$ by:

$$d(x, y) = \frac{|x - y|}{1 + |x - y|}$$

Prove that d is a metric on $\mathbb{R}$. (*Hint:* For the triangle inequality, note that $d(x, y) = f(|x - y|)$ and $f(u) = \frac{u}{1+u}$ is an increasing function.)

Exercise 2.4. Let $X = \mathbb{R}$ and define $d : X \times X \rightarrow [0, \infty)$ by:

$$d(x, y) := |x^n - y^n|$$

Show for which $n \in \mathbb{N}$ d is a metric on $\mathbb{R}$.

Exercise 2.5. Prove that the pair $(\mathcal{C}([a, b]), d_\infty)$ in **Example 2.3** is a metric space.

2.2 Analysis of Metric Spaces

The topology of a metric space is fundamentally determined by specifying which subsets are considered *open* and which are *closed*. These designations capture essential properties needed for rigorous definitions of limits, continuity, and other fundamental concepts in analysis.

Now, we define two special sets.

Definition 2.2 (Open and Closed Balls). Let (X, d) be a metric space. For $x \in X$ and $r > 0$, the *open ball* with center x and radius r is given by

$$B_r(x) = \{y \in X \mid d(x, y) < r\},$$

and the *closed ball* with center x and radius r is given by

$$\overline{B}_r(x) = \{y \in X \mid d(x, y) \leq r\}.$$

We now define open and closed sets.

Definition 2.3 (Open and Closed Sets). Let (X, d) be a metric space. The set $A \subseteq X$ is *open* if for every $x \in A$, there exists $\epsilon > 0$ such that $B_\epsilon(x) \subseteq A$. The set A is *closed* if the complement $A^c = X \setminus A$ is open.

Definition 2.4 (Interior, Exterior, Boundary, and Closure). Let (X, d) be a metric space, $A \subseteq X$ and $x \in X$.

- (a) A point x is an *interior point* of A if there exists $\epsilon > 0$ such that $B_\epsilon(x) \subseteq A$. The set of all interior points of A is denoted by $\text{int}(A)$.
- (b) A point x is an *exterior point* of A if there exists $\epsilon > 0$ such that $B_\epsilon(x) \subseteq A^c$. The set of all exterior points of A is denoted by $\text{ext}(A)$.

- (c) A point x is a *boundary point* of A if for any $\epsilon > 0$, the open ball $B_\epsilon(x)$ has non-empty intersection with both A and A^c , i.e., $B_\epsilon(x) \cap A \neq \emptyset$ and $B_\epsilon(x) \cap A^c \neq \emptyset$. The set of all boundary points of A is denoted by $bd(A)$.
- (d) A point x is a *closure point* of A if for any $\epsilon > 0$, the open ball $B_\epsilon(x)$ has a non-empty intersection with A , i.e., $B_\epsilon(x) \cap A \neq \emptyset$. The set of all closure points of A is denoted by $\bar{A}$.

2.2.1 Open Sets

We will now practice how to prove set structures using the previous definitions.

Exercise 2.6. Let (X, d) be a metric space, $x \in X$ and $r > 0$. Show that the open ball $B_r(x)$ is an open set.

(*Hint:* For any $y \in B_r(x)$, you need to find $s > 0$ such that $B_s(y) \subset B_r(x)$. Consider the geometric relationship between the distances $d(x, y)$ and r to find a proper $s > 0$, and use the triangle inequality to prove it.)

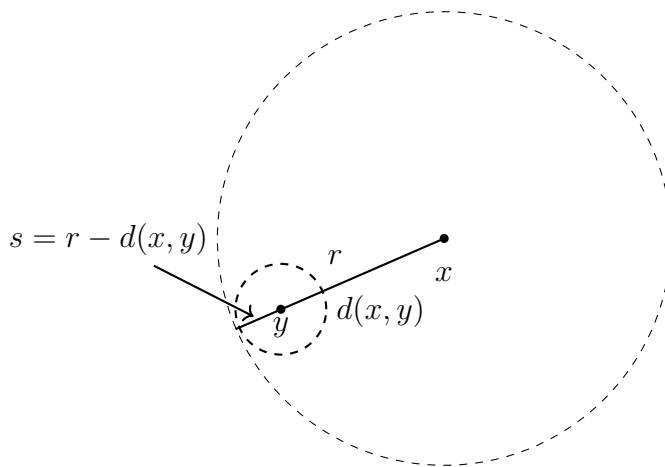


Figure 1: $B_r(x)$

Proof Strategy: In order to prove that $B_r(x)$ is open, we need to prove that, for any $y \in B_r(x)$, there exists some $s > 0$ such that $B_s(y) \subset B_r(x)$. Recall that, for arbitrary sets A and B , we have:

$$A \subset B \iff (x \in A \implies x \in B) \text{ for all } x \in A$$

Therefore, we need to prove that given $y \in B_r(x)$, there exists some $s > 0$ such that $z \in B_s(y) \implies z \in B_r(x)$ for all z .

Figure 1 suggests that $s = r - d(x, y)$ is a good candidate. We will use the triangle inequality to verify that this choice works.

Proof. Let $y \in B_r(x)$ and $s := r - d(x, y)$. Since $y \in B_r(x)$, we have $d(x, y) < r$, so clearly $s > 0$. Consider the open ball $B_s(y)$ with center at y and radius s , and let $z \in B_s(y)$. Then, by the triangle inequality:

$$\begin{aligned} d(x, z) &\leq d(x, y) + d(y, z) \\ &< d(x, y) + s \\ &= d(x, y) + r - d(x, y) \\ &= r \end{aligned}$$

Given that $d(x, z) < r$, we have $z \in B_r(x)$. Hence, $z \in B_s(y) \implies z \in B_r(x)$, which verifies that there exists $s > 0$ such that $B_s(y) \subset B_r(x)$ for all $y \in B_r(x)$. Therefore, $B_r(x)$ is open. $\square$

Exercise 2.7. Show that $A := \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 x_2 \neq 0\}$ is open in $\mathbb{R}^2$. (*Hint:* Think geometrically to find a proper $r > 0$ and then prove this claim.)

Exercise 2.8. Let $a > 0$. Prove that $A := \{(x, y) \in \mathbb{R}^2 \mid x \geq 0, 0 \leq y \leq a\}$ is closed in $\mathbb{R}^2$. (*Hint:* Think geometrically. You need to prove that A^c is open.)

Exercise 2.9. Prove that $A := \mathbb{R} \times \mathbb{Z}$ is closed in $\mathbb{R}^2$. (*Hint:* Think geometrically. Show that A^c is open by writing it as an infinite union of open sets (which sets?) and use the fact that a union of open sets is open.)

Exercise 2.10. Let $A := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \neq 1\}$. Show that A is open in $\mathbb{R}^2$. (*Hint:* Think geometrically. For the formal proof, remember the reverse triangle inequality of the euclidean norm: $|||x|| - ||y||| \leq ||x - y||$ for some $x, y \in \mathbb{R}^n$)

Exercise 2.11. Let $A := \{(x, y) \in \mathbb{R}^2 : 0 \leq x < 1, 0 \leq y < 1\}$. Is A open? (*Hint:* Again, think geometrically. Will be pretty clear.)

Exercise 2.12. Let $A := \{(x, y) \in \mathbb{R}^2 : |x| + |y| < 1\}$. Is A open? (*Hint:* Think geometrically.)

2.2.2 Convergence in Metric Spaces

Before we jump into closed sets, we need to mention some significant definitions regarding convergence in metric spaces.

Definition 2.5 (Convergence in Metric Spaces). Let (X, d) be a metric space, and $\{x_n\}$ a sequence in X . We say that $\{x_n\}$ converges to $x^* \in X$, or that the sequence has limit x^* , if:

$$\forall \epsilon > 0, \exists N_\epsilon \text{ such that } d(x_n, x^*) < \epsilon \quad \forall n > N_\epsilon$$

Definition 2.6 (Limit points). Let (X, d) be a metric space and A be a set in X . A point $x \in X$ is a *limit point* (or *cluster point*) of A if every open ball around it contains at least one point of A , which is distinct from x . Formally, x is a limit point if and only if for each $\epsilon > 0$ we have $B_\epsilon(x) \cap (A \setminus \{x\}) \neq \emptyset$.

The set of all limit points of A is called its **derived set** and is denoted by $D(A)$.

Proposition 2.1. *Let (X, d) be a metric space and A be a set in X . A point $x \in X$ is in $D(A)$ if and only if there exists a sequence in $A \setminus \{x\}$ that converges to x .*

2.2.3 Closed Sets

We have seen in **Definition 2.3** that a set $A \subseteq X$ in the metric space (X, d) is closed if its complement is open. Therefore, in order to prove that A is closed, it is enough to prove that A^c is open, which we have done numerous times above.

Nonetheless, the following results are useful as well (especially **Theorem 2.2 (c)**):

Theorem 2.1. *Let (X, d) be a metric space and A be a set in X . Then the following three statements are equivalent:*

- (a) A is closed.
- (b) A contains all its limit points, i.e., $D(A) \subseteq A$.
- (c) Every convergent sequence in A has its limit in A , i.e., if $x_n \in A$ for all n and $\{x_n\}$ converges to x , then $x \in A$.

That is, in order to prove that $A \subseteq X$ is closed, we can prove that every convergent sequence contained in A has its limit in A , i.e.: if $\{x_n\}$ is a convergent sequence such that $x_n \in A \quad \forall n \in \mathbb{N}$ and $\{x_n\} \rightarrow x^*$, then $x^* \in A$.

2.2.4 Compact Sets

Definition 2.7 (Bounded Set). A subset S of X is *bounded* if there exist $x \in X$ and $r > 0$ such that $S \subset B_r(x)$.

Theorem 2.2 (Heine-Borel). *Consider the metric space $(\mathbb{R}^n, d)$. A set $A \subseteq \mathbb{R}^n$ is compact iff it is closed and bounded.*

Exercise 2.13. Let $H_+(x) := \{x \in \mathbb{R}_+^n \mid a \cdot x \leq b\}$ with $a \in \mathbb{R}_{++}^n$ and $b > 0$. Prove that $H(x)$ is compact.

(Hint: For closedness, use convergent sequences).

3 Convex Analysis

3.1 Convex Sets

Definition 3.1 (Convex Set). A set X is **convex** if given any two points x' and x'' in X , the point

$$x^\lambda = (1 - \lambda)x' + \lambda x''$$

is also in X for every $\lambda \in [0, 1]$.

That is, a set X is convex if all the points in the line segment between any two points in X lie in X .

Example 3.1. Following are examples of convex sets:

- (a) The disc $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq r\}$ is a convex set.
 - (b) The circle $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = r\}$ is **not** a convex set.
 - (c) The hyperplane $H(p, \alpha) = \{x \in \mathbb{R}^n : p \cdot x = \alpha \text{ for } p \in \mathbb{R}^n \text{ and } \alpha \in \mathbb{R}\}$ is a convex set.
-

Exercise 3.1 (Open ball). Consider the Euclidean metric space $(\mathbb{R}^n, \|\cdot\|)$. Let $x \in \mathbb{R}^n$ and $r > 0$. Prove that the open ball $B_r(x) = \{y \in \mathbb{R}^n : \|x - y\| < r\}$ is convex.

Proof. We need to show that for any two points $y_1, y_2 \in B_r(x)$ and any $\lambda \in [0, 1]$, the convex combination $y^\lambda = (1 - \lambda)y_1 + \lambda y_2$ is also in $B_r(x)$.

Let $y_1, y_2 \in B_r(x)$. Then by definition: $\|x - y_1\| < r$ and $\|x - y_2\| < r$.

Consider the convex combination $y^\lambda = (1 - \lambda)y_1 + \lambda y_2$ for any $\lambda \in [0, 1]$.

We need to show that $y^\lambda \in B_r(x)$, i.e., that $\|x - y^\lambda\| < r$:

$$\begin{aligned}
 \|x - y^\lambda\| &= \|x - ((1 - \lambda)y_1 + \lambda y_2)\| \\
 &= \|x - (1 - \lambda)y_1 - \lambda y_2\| \\
 &= \|\lambda x - \lambda x + x - (1 - \lambda)y_1 - \lambda y_2\| \\
 &= \|(1 - \lambda)x + \lambda x - (1 - \lambda)y_1 - \lambda y_2\| \\
 &= \|(1 - \lambda)(x - y_1) + \lambda(x - y_2)\| \\
 &\leq \|(1 - \lambda)(x - y_1)\| + \|\lambda(x - y_2)\| \quad (\text{triangle inequality of norms}) \\
 &= (1 - \lambda)\|x - y_1\| + \lambda\|x - y_2\| \quad (\text{homogeneity of norms})
 \end{aligned}$$

Since $y_1, y_2 \in B_r(x)$, we have $\|x - y_1\| < r$ and $\|x - y_2\| < r$. Then:

$$\begin{aligned}
 \|x - y^\lambda\| &\leq (1 - \lambda)\|x - y_1\| + \lambda\|x - y_2\| \\
 &< (1 - \lambda)r + \lambda r \\
 &= r
 \end{aligned}$$

Therefore, $\|x - y^\lambda\| < r$, which means $y^\lambda \in B_r(x)$. Hence, $B_r(x)$ is convex. $\square$

Exercise 3.2 (Solution set of linear equations). Let $C := \{x \in \mathbb{R}^n \mid Ax = b\}$ be the set of all solutions to the system of linear equations $Ax = b$, where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Show that C is convex. (*Hint: Take two elements from C and show that $x^\lambda \in C$ algebraically.*)

Proof. We need to show that for any two points $x_1, x_2 \in C$ and any $\lambda \in [0, 1]$, the convex combination $x^\lambda = (1 - \lambda)x_1 + \lambda x_2$ is also in C .

Let $x_1, x_2 \in C$. Then by definition of C : $Ax_1 = b$ and $Ax_2 = b$. Next, consider the convex combination $x^\lambda = (1 - \lambda)x_1 + \lambda x_2$ for any $\lambda \in [0, 1]$.

We need to show that $x^\lambda \in C$, i.e., that $Ax^\lambda = b$:

$$\begin{aligned}
 Ax^\lambda &= A[(1 - \lambda)x_1 + \lambda x_2] \\
 &= A(1 - \lambda)x_1 + A\lambda x_2 \quad (\text{linearity of matrix multiplication}) \\
 &= (1 - \lambda)Ax_1 + \lambda Ax_2 \quad (\text{distributivity}) \\
 &= (1 - \lambda)b + \lambda b \quad (\text{since } Ax_1 = b \text{ and } Ax_2 = b) \\
 &= b[(1 - \lambda) + \lambda] \\
 &= b
 \end{aligned}$$

Therefore, $Ax^\lambda = b$, which means $x^\lambda \in C$. Hence, the set C is convex. $\square$

Exercise 3.3. Let $a > 0$. Prove that $A := \{(x, y) \in \mathbb{R}^2 \mid x \geq 0, 0 \leq y \leq a\}$ is convex. (*Hint: Take two elements from A and show that $x^\lambda \in A$ algebraically.*)

Exercise 3.4. Prove **Example 3.1 (a)**. (*Hint: Think geometrically, then take two points from D and show that $x^\lambda \in D$ algebraically. At some point, the Cauchy-Schwarz inequality maybe useful: $x_1x_2 + y_1y_2 \leq \sqrt{x_1^2 + y_1^2}\sqrt{x_2^2 + y_2^2}$*)

Exercise 3.5. Prove **Example 3.1 (b)**. (*Hint: Think geometrically and find a counterexample.*)

Exercise 3.6. Let X be a convex set in $\mathbb{R}^n$, and let α be a real number. Show that the set

$$\alpha X = \{z \in \mathbb{R}^n : z = \alpha x \text{ for some } x \in X\}$$

is convex. (*Hint: Take two elements from αX and show that $x^\lambda \in \alpha X$ algebraically.*)

Exercise 3.7. Let X and Y be convex sets in $\mathbb{R}^n$. Show that the set

$$X + Y = \{z \in \mathbb{R}^n : z = x + y \text{ for some } x \in X \text{ and } y \in Y\}$$

is convex. (*Hint: Take two elements from $X + Y$ and show that $x^\lambda \in X + Y$ algebraically.*)

Exercise 3.8. Let $C(\mathbb{R})$ be the vector space of continuous real-valued functions. Let $\mathcal{D} \subset C(\mathbb{R})$ be the set of those f such that

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad \text{and} \quad f(x) \geq 0 \text{ for all } x \in \mathbb{R}$$

Show that $\mathcal{D}$ is convex.

(*Hint: Use the linearity of integration to show that convex combinations of this type of functions are also in $\mathcal{D}$.*)

4 Concave and Quasiconcave Functions

Theorem 4.1. *A function $f : X \rightarrow \mathbb{R}$ is concave (convex) on X if and only if for all $x, y \in X$ and for all $\lambda \in [0, 1]$, it is the case that*

$$f(\lambda x + (1 - \lambda)y) \geq (\leq) \lambda f(x) + (1 - \lambda)f(y)$$

Theorem 4.2. *Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be two concave functions. Then*

(a) *the function $\alpha f + \beta g : \mathbb{R}^n \rightarrow \mathbb{R}$ is concave for $\alpha, \beta \geq 0$,*

(b) *the function $\min\{f, g\} : \mathbb{R}^n \rightarrow \mathbb{R}$ is concave.*

Exercise 4.1. Prove Theorem 4.2.

Proof. We prove part (a) and leave the rest for practice.

(a): Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be concave functions and $\alpha, \beta \geq 0$. Define $h(x) := \alpha f(x) + \beta g(x)$. Let $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$.

Then

$$h(\lambda x + (1 - \lambda)y) = \alpha f(\lambda x + (1 - \lambda)y) + \beta g(\lambda x + (1 - \lambda)y)$$

Since f and g are concave:

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &\geq \lambda f(x) + (1 - \lambda)f(y) \\ g(\lambda x + (1 - \lambda)y) &\geq \lambda g(x) + (1 - \lambda)g(y) \end{aligned}$$

And given that $\alpha, \beta \geq 0$, we have then:

$$\begin{aligned} h(\lambda x + (1 - \lambda)y) &= \alpha f(\lambda x + (1 - \lambda)y) + \beta g(\lambda x + (1 - \lambda)y) \\ &\geq \alpha[\lambda f(x) + (1 - \lambda)f(y)] + \beta[\lambda g(x) + (1 - \lambda)g(y)] \\ &= \lambda[\alpha f(x) + \beta g(x)] + (1 - \lambda)[\alpha f(y) + \beta g(y)] \\ &= \lambda h(x) + (1 - \lambda)h(y) \end{aligned}$$

Therefore, h is concave. □

Theorem 4.3. *A function $f : X \rightarrow \mathbb{R}$ is quasiconcave on X if and only if for all $x, y \in X$ and*

for all $\lambda \in [0, 1]$, it is the case that

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}.$$

Similarly, a function $f : X \rightarrow \mathbb{R}$ is quasiconvex on X if and only if for all $x, y \in X$ and for all $\lambda \in [0, 1]$, it is the case that

$$f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}.$$

Theorem 4.4. *Let $f : X \rightarrow \mathbb{R}$ be concave on X , then it is also quasiconcave. Similarly, if $f : X \rightarrow \mathbb{R}$ is convex on X , then it is also quasiconvex.*

Theorem 4.5. *Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be two quasiconcave functions. Then*

- (a) *the function $\alpha f : \mathbb{R}^n \rightarrow \mathbb{R}$ is quasiconcave for $\alpha \geq 0$,*
- (b) *the function $\min\{f, g\} : \mathbb{R}^n \rightarrow \mathbb{R}$ is quasiconcave,*
- (c) *if $h : \mathbb{R} \rightarrow \mathbb{R}$ is non-decreasing, then $h \circ f : \mathbb{R}^n \rightarrow \mathbb{R}$ is quasiconcave.*

Exercise 4.2. Prove Theorem 4.5.

Proof. We prove part (a) and leave the rest for practice.

(a): Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be quasiconcave and $\alpha \geq 0$. Define $h(x) := \alpha f(x)$. Let $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$.

Since f is quasiconcave and $\alpha \geq 0$:

$$\begin{aligned} h(\lambda x + (1 - \lambda)y) &= \alpha f(\lambda x + (1 - \lambda)y) \\ &\geq \alpha \min\{f(x), f(y)\} \\ &= \min\{\alpha f(x), \alpha f(y)\} \\ &= \min\{h(x), h(y)\} \end{aligned}$$

Therefore, h is quasiconcave. □

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